7th

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a priori error estimation:

$$\begin{cases} -(au')' = f & \text{in } (0,1) \\ u(0) = u(1) = 0 \end{cases}$$

$$V = \left\{ v : \int_0^1 [v^2 + (v')^2]\,dx < \infty, \; v(0) = v(1) = 0 \right\}$$

V_h for CG(1).

(1) energy norm:

$$\|e\|_E^2 = \int_0^1 a[(u-U)']^2\,dx$$

Last time it was shown that $\Rightarrow$ FEM

$$\|u-U\|_E \le \|u-v\|_E \quad \forall v \in V_h$$

(any other function in V_h)

Now, take the test function

$$v = \Pi_h u \in V_h$$

projection of $u \in V$ into the subspace $V_h \subset V$

Here,

$$\Pi_h u = \sum_{j=1}^{m} u(x_i)\,\phi_i(x)$$

↑ interpolate $u(x)$ onto the mesh using piecewise Green. functions.

↑ continuous.

Note:

$$\|u - \Pi_h u\|_E = \|(u - \Pi_h u)'\|_a \le C\,\|h u''\|_a$$

This just come from approximating u' by a piecewise constant function.

C is a constant that only depends on $a(x)$.

Thm: The FEM approx. $U \in V_h$ satisfies

$$\|u - U\|_E \le \|u - v\|_E \quad \forall\ v \in V_h \subset V$$

In particular, there exist a constant C, depending only on $a(x)$. such that

$$\|u-U\|_E \le C\|hu''\|_a$$

Note: this shows that $eG(1)$ is first order accurate in the energy norm, provided that $U'' \in L^\infty(0,1)$.

(2) $L^\infty(0,1)$ norm:

$$\begin{cases} -(au')' = f & \text{in } 0,1 \\ u(0) = u(1) = 0 \end{cases}$$

consider the following dual problem:

$$\left.\begin{aligned} -(a\phi')' &= e \\ \phi(0) = \phi(1) &= 0 \end{aligned}\right\} \text{ here } e = u - U$$

Note: because the original 2-pt BVP is self adjoint, the same.

self adjoint,

primal & dual operators are the same.

L^2-norm:

$$\|e\|^2 = \int_0^1 e^2 \, dx$$

$$= -\int_0^1 e\,(a\phi')'\,dx$$

$$= \int_0^1 a e' \phi' \, dx - [a e \phi']_0^1 \quad (\to 0)$$

but $e(0) = e(1) = 0$

$$\|e\|^2 = \int_0^1 a e' \phi' \, dx$$

Recall: Galerkin orthogonality:

$$\int_0^1 a e' v' \, dx = 0 \qquad \forall\, v \in V_h$$

take $v = \pi_h \phi$

$$\|e\|^2 = \int_0^1 a e' \phi' \, dx - \int_0^1 a e' (\pi_h \phi)' \, dx$$

$$= \int_0^1 a e' (\phi - \pi_h \phi)' \, dx$$

cauchy inequality :

$$\|e\|^2 \leq \|e''\|_a \, \|(\phi - \pi_h \phi)'\|_a$$

$$\|e\|^2 \leq C_I h_{max} \|e'\|_a \underbrace{\|\phi''\|_a}_{\text{interpolation error}}$$

but recall :

$$\|e'\|_a = \|e\|_E \leq C_I h_{max} \|u''\|_0$$

↳ From part (1)

$$\|e\|^2 \leq C_I^2 h_{max}^2 \|u''\|_a \|\phi''\|_a$$

$$\|e\|^2 \leq C_I^2 h_{max}^2 \|u''\|_a \frac{\|\phi''\|_a}{\|e\|} \|e\|$$

$$\|e\| \leq C_I^2 h_{max}^2 \|u''\|_a \frac{\|\phi''\|_a}{\|e\|}$$

let, $S_a = \max_{\xi \neq 0} \frac{\|\psi''\|_a}{\|\xi\|}$

where $-(a(\psi)')' = \xi$

where, $-(a\varphi')' = \zeta$

$\varphi(0) = \varphi(1) = 0$

Note: if $a \equiv$ constant

$-(a\varphi')' = -a\varphi'' = \zeta$

$\Rightarrow \dfrac{\varphi''}{\|\zeta\|} = \dfrac{1}{a}$